

Theoretical Computer Science, exam of 2026.01.14

SURNAME _____ NAME _____ PERSONAL CODE _____

Total available time: 1h 30'. You can use books and paper notes during the exam, while electronic devices are prohibited. **You must write the solutions on this paper alone.**

Each correct exercise is valued 10 points (+1 bonus for the best solutions).

Multichance students do not need to answer to Exercises 2.3 and 3.2: bar here in that case ☐

Exercise 1

Consider the following language, and provide a recognizer or a grammar of minimal power for it.

$$L = \{a^n b^{n+m+1} a^{3m} \mid n, m \geq 0\}.$$

Exercise 2

State whether the following functions are computable, providing appropriate justification.

1) $g_0(x,y,z) = f_w(f_x(z)) - f_z(x)$, with $w = f_y(z)$.

2) $g_1(x,y,z) = f_z(x)$ if $f_y(x) = 2 = f_x(y)$; $g_1(x,y,z) = \perp$ otherwise.

3) $g_2(x,y,z) = 1$ if $f_y(x) = \perp = f_z(x)$; $g_2(x,y,z) = 2$ otherwise.

Exercise 3

Consider the language L of Exercise 1.

1) Describe a single-tape Turing machine that recognizes the language L , and evaluate its space and time complexity.

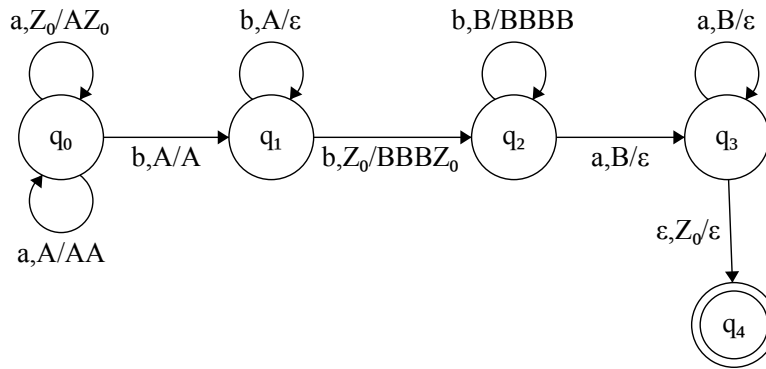
2) Describe a RAM machine that recognizes the language L , and evaluate its space and time complexity.

space available for solutions

Solutions

EX 1

The language L is recognizable by using the following **deterministic pushdown automaton**.



EX 2

- 1) Computable, since f_x , f_y and f_z are computable by definition.
- 2) Computable: we can compute $f_y(x)$ and $f_x(y)$: if both the computations stop, then we check if their values are both equal to 2 and return $f_z(x)$, otherwise we loop.
- 3) Non computable: we can reduce the halting problem function g to g_2 in this way:

$$g(y,x) = 1 + g_2(x,y,y).$$

EX 3

- 1) The TM considers first the first factor of a's:
 - 1) In each pass it replaces each first found a by N, and each first found b by N.
 - 2) It then replaces the next b by N.
 - 3) In each pass it replaces each first found b by M, and each first found aaa by M.
 If the resulting string belongs to N^+M^+ , it accepts.

Complexity: $T(n) = \Theta(n^2) + \Theta(n^2) = \Theta(n^2)$, $S(n) = \Theta(n)$, with n length of the string.

- 2) The RAM machine is straightforward: it keeps three counters x , y , and z , respectively for the first factor of a's, the factor of b's, and then the last a's. It then checks if $y = x + z/3 + 1$ in constant time.
 Complexity:

Constant criterion: $T(n) = \Theta(n)$, $S(n) = \Theta(1)$, with n length of the string.

Log criterion: $T(n) = \Theta(n)$, $S(n) = \Theta(\log n)$.